

Jian Wei CHEONG

✉ contact@jianwei.simplelogin.com
🌐 jianweicheong.codeberg.page
ID 0000-0001-7114-7825
🏠 Google Scholar

Curriculum Vitae

Professional Qualifications

- 2019 - 2023 **PhD, Physics**, *Nanyang Technological University (NTU)*, Singapore
- 2015 - 2019 **BSc, Physics, Honours**, *NTU*, Singapore
- 2010 - 2013 **Diploma, Electrical Engineering**, *Ngee Ann Polytechnic (NP)*, Singapore

Professional Experience

- 2024 - current **Postdoctoral Research Fellow**, *NTU*, Singapore
- 2023 **Project Officer**, *NTU*, Singapore
- 2013 - 2015 **Mandatory Military Service**, *Singapore Armed Forces*, Singapore
- 2012 **Intern**, *ST Electronics*, Singapore

Teaching Experience

- 2026/2025/2024 **Computational Nonlinear Lab (Physics Discovery Camp)**, *NTU*, Singapore
- 2025/2021 **PH3101 Quantum Mechanics 2**, *NTU*, Singapore
- 2020 **PH1199 Physics Lab 1B**, *NTU*, Singapore
- 2019 **PH1198 Physics Lab 1A**, *NTU*, Singapore

Awards & Achievements

- 2019 **Short-speech Contest Best Presentation** (PAP701 Graduate seminar module), *NTU*, Singapore
- 2017/2018 **Dean's List** (top 5% of cohort), *NTU*, Singapore
- 2016/2017 **NTU President Research Scholar** (completing URECA), *NTU*, Singapore
- 2011 **Director's List** (top 5% of cohort), *NP*, Singapore
- 2011 **Best Performance, Programmable Logic Device** (top student of cohort), *NP*, Singapore
- 2010 **Best Performance, Digital Electronics & Practice** (top student of cohort), *NP*, Singapore

Bibliography

Publications

1. Y. Ma, [J. W. Cheong](#), K. Huang, H. Liu, X. Long, T. Xin, X. Nie, D. Lu, **Non-Markovianity supplies the magic behind indefinite-causal-order-based quantum refrigeration**, *Physical Review Letters* (2026) (*Under Review*).
2. Y. Wang, A. Pradana, [J. W. Cheong](#), and L. Y. Chew, **Enhancing performance and operational regimes of the quantum Otto refrigerator with heat bath algorithmic cooling**, *Physical Review E*, **113**(5), 054112 (2026).
3. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Enhancing the performance of quantum switch refrigeration with an information pool of qubits**, *Physical Review Research*, **8**(1), 013272 (2026).
4. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Non-Markovian refrigeration and heat flow in the quantum switch**, *Physical Review A*, **110**(2), 022220 (2024).
5. L. Y. Chew, A. Pradana, L. He, and [J. W. Cheong](#), **Stochastic thermodynamics of finite-tape information ratchet**, *European Physical Journal Special Topics* (2023).
6. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Effects of non-Markovianity on daemonic ergotropy in the quantum switch**, *Physical Review A*, **108**(1), 012201 (2023).
7. L. He, [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Effects of correlation in an information ratchet with finite tape**, *Physical Review E*, **107**(2), 024130 (2023).
8. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Communication advantage of quantum compositions of channels from non-Markovianity**, *Physical Review A*, **106**(5), 052410 (2022).
9. L. He, A. Pradana, [J. W. Cheong](#), and L. Y. Chew, **Information processing second law for an information ratchet with finite tape**, *Physical Review E*, **105**(5), 054131 (2022).

Proceedings

1. L. Y. Chew, [J. W. Cheong](#), and A. Pradana, **Thermodynamic Functionality of Non-detailed Balance Finite-Tape Information Ratchet**, *International Conference on Geometric Science of Information, Springer Nature Switzerland*, pp. 173–181 (2025).

Talks

1. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Non-Markovian Quantum Refrigeration in the Quantum Switch**, *Quantum Thermodynamics Conference 2026 (QTD2026)*, Natal, Brazil, August 2026.
2. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Maxwell's Uncertain Demon: Achieving Nonclassical and Enhanced Refrigeration when Causality Blurs**, *Asia-Pacific Conference on Networks and Complex Systems 2026 (APCNCS 2026)*, Singapore, June 2026.

Posters

1. [J. W. Cheong](#), A. Pradana, and L. Y. Chew[†], **Enhancement of quantum processes from indefinite causal order through non-Markovianity**, *29th International Conference on Statistical Physics (STATPHYS29)*, Florence, Italy, July 2025.
2. [J. W. Cheong](#)[†], A. Pradana, and L. Y. Chew, **Non-Markovian refrigeration and heat flow in the quantum switch**, *Quantum Thermodynamics Conference 2025 (QTD2025)*, Singapore, July 2025.
3. L. Y. Chew[†], [J. W. Cheong](#), and A. Pradana, **Enhancement of quantum processes from indefinite causal order through non-Markovianity**, *XLV Dynamics Days Europe 2025 (DDE2025)*, Thessaloniki, Greece, June 2025.
4. L. Y. Chew[†], L. He, A. Pradana, and [J. W. Cheong](#), **Stochastic thermodynamics and correlation effects of finite-tape information ratchets**, *28th International Conference on Statistical Physics (STATPHYS28)*, Tokyo, Japan, August 2023.

[†]Presenting author.

Technical Experience

	Skill	Level	Comment
Research	Non-Markovianity		main research
	Indefinite causality		main research
	Channel compositions		main research
	Refrigeration		application of main research
	Work engines		application of main research
	Quantum communication		application of main research
	Maxwell's demon		main + secondary research
	Information ratchet		secondary research
	Autonomous systems		secondary research
	Quantum battery		secondary research
	Entanglement monogamy		side research
	Resource theory		just starting (collaboration)
	Quantum reference frames		just starting (collaboration)
	NMR		light experience (collaboration)
	NV center magnetometers		light experience
Programming	Python / Julia / R		used in main work
	C / C++		bachelor course, RCpp
	MATLAB		bachelor course
	Haskell / Racket / Lisp		personal projects
	Bash / POSIX sh		personal Linux projects
	Quarto / LaTeX / Typst		presentations, reports
	HTML / CSS		personal website
Misc	Arduino & microcontrollers		bachelor & diploma courses
	Blender		scientific visualization
	LabVIEW		bachelor & diploma courses
		basic knowledge	extensive knowledge
		intermediate knowledge	expert knowledge

Non-Quantum Project Experience

- **Strain estimation for hazard forecastings before and after 2011 Japan Tohoku earthquake** by analysis of seismic GPS displacement data and computation of seismic strains from velocity fields.
ES7008 Geophysical Data Analysis, NTU
- **Variations in statistical complexity of genome sequences across species** by analysis of genome sequences from GenBank assembly with Baum-Welch algorithm.
CE7412 Computational and Systems Biology, NTU
- **Detecting adversarial attack of deep neural networks for image recognition from image complexity** by Fast Gradient Sign Method (FGSM), DeepFool, One Pixel Attack, and Jacobian-Based Saliency Map Attack (JSMA).
PH3502 Chaotic Dynamical Systems, NTU
- **Monte Carlo photon transport in multi-layered biological tissues** with Henyey-Greenstein scattering for diagnostic imaging and photon therapy.
PH4505 Computational Physics, NTU
- **Monte Carlo simulation of periodic-driven Brownian particles** with Arrhenius equation, demonstrating dissipation-driven adaptation.
PAP723 Numerical Methods for Physicists, NTU